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Multiphysics analysis of thermal mechanical behavior and tritium migration in FeCrAl and Cr-coated Zircaloy cladding under PWR normal operating and transient conditions

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ABSTRACT

Both Cr-coated Zircaloy and the FeCrAl alloy have garnered attention as potential substitutes for PWR cladding materials, due to their impressive qualities, such as robust high-temperature oxidation and corrosion resistance. However, FeCrAl alloy is found to accelerate the tritium permeation throughout the cladding, this heightened tritium permeation prompts easier release from the cladding into the coolant. Consequently, the assessment of material tritium permeation becomes a crucial aspect in fuel performance analysis. In this work, the multiphysics models of Cr-coated Zircaloy and FeCrAl cladding are reviewed firstly. Then the multiphysics models are implemented in CAMPUS code and the validation of tritium migration model is presented. Subsequently, the fuel performance under normal and accident (LOCA and RIA) conditions and the tritium migration in three types of claddings are simulated and discussed. The simulation results show that the utilization of FeCrAl cladding, instead of Zircaloy, results in an overall reduction in fuel temperature under both normal and LOCA conditions. Furthermore, both the Cr-coated Zircaloy and FeCrAl cladding are found to effectively delay the cladding failure time under LOCA condition. However, the total strain of FeCrAl cladding is found to be significantly higher than that of the Cr-coated Zircaloy cladding under RIA condition. And the Cr-coated Zircaloy is found to have better tritium resistance effect than FeCrAl cladding, which greatly reduces the coolant contamination treatment. Therefore, based on the combined consideration of fuel performance and tritium resistance, Cr-coated Zircaloy emerges as the preferred candidate cladding over the FeCrAl cladding.

1. Introduction

Following the Fukushima nuclear accident in 2011, accident tolerant fuel (ATFs) have been proposed as replacement materials in nuclear reactors, since ATFs are intended to enhance the safety and economy of nuclear reactor operation. The Zircaloy cladding is suggested to be substituted by the ATF materials since Zircaloy is found to react exothermically with high-temperature water and vapor under accident conditions, and therefore generate a large amount of hydrogen and heat from the oxidization process, which can lead to an explosion risk in the presence of air. Recently, efforts have been made to explore the performance of ATF materials. The common ideas in the development of ATF cladding materials have been proposed (Kim et al., 2016), which are divided into long-term and short-term goals. The long-term objective revolves around the creation of novel materials with robust strength and heightened oxidation resistance, FeCrAl alloys (Terrani et al., 2014), SiC composites (Yueh and Terrani, 2014), and Mo alloys (Cheng et al., 2016) have been studied most for this purpose. Conversely, the short-term goal entails the enhancement of the existing Zircaloy by concentrating on its high-temperature properties, particularly at 1200 °C. This can be achieved by applying surface coating protection, enabling optimal performance at elevated temperatures, while maintaining the fundamental design of the current reactor (Tang et al., 2017).

Recent developments have introduced a variety of coating options for cladding surfaces, encompassing carbide, nitride, alloy coatings, and more. Among these options, the concept of Zircaloy cladding with protective coatings has emerged (Kashkarov et al., 2021) as a means to enhance fuel safety during both normal operations and accident scenarios. However, certain coatings have demonstrated a dual nature: while displaying favorable characteristics in high-temperature oxidation, they also exhibit vulnerability to cracking due to brittleness and thermal expansion differences between Zircaloy and coatings, such as

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SiC coating and CrN coating. FeCrAl alloy is also proposed to be used as one of the coatings based on Zircaloy (Han, 2019), Yet, concerns arise from observed interdiffusion between FeCrAl and Zircaloy, leading to the formation of a problematic Fe–Zr eutectic phase and subsequent failure of the FeCrAl coating. In contrast, Cr coating is discovered to tend to form a dense Cr_2O_3 film on the cladding surface, which is found to have excellent chemical stability, high temperature oxidation resistance. And attention has been attracted by the Cr material as a potential coating on Zircaloy under accident conditions, to limit external cladding corrosion in high temperature steam (Yeom et al., 2019). Furthermore, the commendable attributes of Cr include a high melting point, robust corrosion resistance, and a thermal expansion coefficient akin to that of Zircaloy (Brachet et al., 2019). The utilization of Cold spraying technology, as demonstrated by Park et al. (2016), facilitated the deposition of Cr coatings onto Zircaloy plates. Through the LOCA condition test, the Cr-coated Zircaloy is identified to have smaller hoop strain and smaller fracture size. Kim et al. (2015) observed that the oxide layer thickness formed on the Cr-coated tube surface is about 25-times thinner compared to Zircaloy-4 tube surfaces, achieved using 3D laser coating technology with Cr powder as the coating material. Collectively, these studies underscore the advantageous role of Cr-coated Zircaloy cladding in preserving fuel cladding integrity and overall safety.

Furthermore, among the promising directions in the research on ATF cladding, the FeCrAl alloy cladding emerges as a noteworthy candidate for potentially replacing Zircaloy. After rigorous testing and comprehensive evaluation, the FeCrAl alloy has demonstrated remarkable corrosion resistance in typical operational conditions, leading to enhanced fuel performance compared to traditional Zircaloy cladding (Qiu et al., 2018). Rebak (2017) emphasized that not only does FeCrAl alloy cladding prevent cladding failures, but it also strengthens the cladding when employed in lieu of UO_2-Zircaloy under accident conditions. There is a focus on FeCrAl alloy, which exhibits much slower oxidation kinetics in high-temperature steam than Zircaloy. The cladding is allowed to remain for longer periods of time in high temperature steam, which makes accident mitigation more likely (Sweet et al., 2018). In efforts to comprehensively assess the performance of FeCrAl alloy under both pressurized water reactor (PWR) normal operating conditions and accident scenarios, Gamble (2017) updated the fuel performance analysis code BISON. Their simulations revealed that the behavior of FeCrAl alloy closely resembles that of Zircaloy, with a notable exception being its markedly improved oxidation properties—surpassing those of Zircaloy cladding by a significant margin. Moreover, Gamble (2017) pointed out a delayed rupture time for FeCrAl cladding under high temperatures. On the other hand, compared with the existing Zircaloy, hydrogen generation is reduced due to the improved oxidation resistance by using advanced iron alloys, which would favorably contribute to increased coping time in the case of accidents exceeding the design basis (Terrani et al., 2014).

While FeCrAl alloy offers the advantage of forming a thin aluminum oxide film on its surface at high temperatures, providing protection against further oxidation in steam conditions (Kim and Rebak, 2016), it also presents a potential drawback in terms of promoting tritium penetration. Tritium, a notable radionuclide, is generated in significant quantities within nuclear power plants and constitutes a substantial portion of the radioactive released into the environment from such facilities. Notably, in pressurized water reactors, tritium is produced through ternary fission of uranium nuclei. This tritium then permeates the coolant via the cladding, contributing approximately 20% of the overall tritium content within the coolant. In accordance with the national standard GB/6249-2011, which pertains to a 3000 MW light water reactor, specific limits have been established for annual emissions of gaseous and liquid tritium. These limits are set at no more than 1.5×10^{13} Bq and 7.5×10^{13} Bq, respectively (Chen et al., 2016). The potential risks associated with tritium stem from its extended half-life and its rapid isotope exchange rate. This characteristic makes tritium readily disperse into the environment in the forms of HTO and HT, giving rise to unfavorable effects. Hence, an imperative exists to investigate tritium permeation behavior within cladding materials to minimize its diffusion into the coolant. Research conducted by Hu et al. (2015) explored the impact of various cladding materials on tritium diffusion, revealing that Cr material exhibits lower tritium permeability than FeCrAl alloy. Additionally, an effective tritium barrier-zirconia inside and outside is formed after using the Zircaloy cladding, which significantly diminishes tritium permeation compared to FeCrAl alloy. It's worth noting that the higher diffusion coefficient of hydrogen in FeCrAl alloy, in contrast to Zircaloy, leads to easier release of tritium produced by pellets from the FeCrAl alloy cladding into the coolant, raising potential safety concerns (Ting et al., 2017). However, the tritium diffusion was found to be reduced after coating a stable Al_2O_3 film on the steel surface (Garud et al., 2022; Garud and Rebak, 2023). Moreover, Feng et al. (2018) successfully developed a composite coating of Al_2O_3/Cr_2O_3, serving as a potential tritium permeation barrier on steel surfaces. This coating significantly enhances the resistance of FeCrAl cladding against tritium permeation.

Currently, few reports have been studied on considering tritium migration properties while analyzing the fuel performance for ATF materials. Therefore, when comparing different new ATF materials, it is important to evaluate not only their fuel performance under different operating conditions, but also to scrutinize their impact on tritium migration. To address this, a multi-scale and multi-physics finite element analysis modeling method is employed in this work. Based on the fuel performance analysis code CAMPUS, the multi-physics models related to the Cr-coated Zircaloy cladding and FeCrAl alloy cladding are updated and analyzed in COMSOL. Besides, we extend the tritium migration module on the established model and the tritium resistance properties of different claddings are further calculated and discussed.

2. Materials and methods

The comprehensive material properties of UO_2 pellets and Zircaloy cladding have been introduced in detail in our previous work of Liu et al. (2016). And the related material properties of the updated Cr coating and FeCrAl cladding will be mainly introduced in this paper.

2.1. Material properties of pure Cr

2.1.1. Thermal conductivity

According to the study of Holzwarth and Stamm (2002), the thermal conductivity of chromium coating in the temperature range of 300K–1300K can be calculated by the following equation:

$$k(T) = -2.07 \times 10^{-8}T^3 + 4.85 \times 10^{-5}T^2 - 0.06T + 101.75 \tag{1}$$

where k is thermal conductivity in $W/(m \cdot K)$ and T is temperature in K.

2.1.2. Heat capacity

Furthermore, as suggested by Holzwarth and Stamm (2002), the expression for the heat capacity in the effective temperature range of 200K–2400K is as follows:

$$C_P(T) = -1.28 \times 10^{-7}T^3 + 3.39 \times 10^{-4}T^2 - 0.09T + 483.2 \tag{2}$$

where C_P is heat capacity in $J/(kg \cdot K)$ and T is temperature in K.

2.1.3. Young's modulus

The Young's modulus is calculated as a function of temperature recommended by Armstrong and Brown (1964):

$$E(T) = -2.50 \times 10^{-5}T^2 - 0.01T + 264.11 \tag{3}$$

where E is Young' s modulus in GPa and T is temperature in K.

2.1.4. Poisson's ratio

The Poisson's ratio with a value of 0.22 proposed by Simmons (1965) of the chromium coating is adopted.

The Cr layer examined in this study is assumed to be uniformly distributed across the cladding's surface. Additionally, the thickness of the Cr coating applied to the Zircaloy cladding is notably thin. It's worth noting that Wagih et al. (2018) indicated that the fuel's thermo-mechanical behavior in the case of Cr-coated Zircaloy cladding is nearly identical to that of uncoated Zircaloy cladding. Therefore, in this work, the equivalently homogenized mechanical properties are used to evaluate the deformation and displacement of the Cr-coated Zircaloy cladding. For the assessment of the entire Cr-coated Zircaloy cladding, the creep model established for the Zircaloy cladding is utilized.

2.2. Material properties of FeCrAl

The material properties of FeCrAl cladding mainly suggested by Field et al. (2018) are summarized below.

2.2.1. Thermal conductivity

The thermal conductivity formula for FeCrAl cladding is applied in the temperature range of 300K to 1400K:

$$k(T) = -7.223\times10^{-7}T^2 + 1.563\times10^{-2}T + 6.569 \tag{4}$$

2.2.2. Heat capacity

The equation for the heat capacity of FeCrAl cladding in a specific temperature range of 300K to 852K is provided below:

$$C_P = 2.982\times10^{-6}T^3 - 4.311\times10^{-3}T^2 + 2.54T \tag{5}$$

2.2.3. Young's modulus

$$E(T) = -5.46\times10^{-5}T^2 - 3.85\times10^{-2}T + 199 \tag{6}$$

2.2.4. Thermal expansion strain

The thermal expansion strain formula of FeCrAl cladding in the effective temperature range of 300 K to 1733 K is as follows:

$$\frac{\Delta L}{L(293K)} = -1.938\times10^{-3} + 3.450\times10^{-6}T + 1.080\times10^{-8}T^2 \tag{7}$$

2.2.5. Creep models

In this paper, the thermal creep and irradiation creep are considered for FeCrAl cladding. Norton's creep law formula proposed by Saunders et al. (1997) about the thermal creep is adopted:

$$\dot{\varepsilon}_{c,th} = C \bullet \sigma^n \bullet exp\left(-\frac{Q}{RT}\right) \tag{8}$$

where C is the creep exponent with a value of 5.96×10^{-24}, σ is the effective stress in Pa, n is a stress fitting parameter with a value of 5.5, and Q is the creep activation energy with a value of $3.92\times10^5 \mathrm{J\cdot mol^{-1}}$, T is the temperature in K, and R is the gas constant in $\mathrm{J\cdot mol^{-1}\cdot K^{-1}}$.

The expression for the irradiation creep of FeCrAl cladding proposed by Karlsen (2015) is as follows:

$$\dot{\varepsilon}_{c,irr} = 4.5\times10^{-32}\sigma\varphi \tag{9}$$

where σ is the effective stress in MPa, φ is the fast neutron flux with a value of $9.5\times10^{17}\ \mathrm{n/m^2\cdot s}$.

2.2.6. Poisson's ratio

The generalized empirical relationship with respect to temperature can be described as follows:

$$v = 4.46\times10^{-5}T + 0.27 \tag{10}$$

2.2.7. FeCrAl cladding failure criterion

The assessment of FeCrAl cladding failure revolves around evaluating whether the hoop stress exceeds the material's ultimate tensile strength (UTS). The UTS, as indicated by Yamamoto et al. (2015) 's experimental findings, is considered as a function of temperature. This functional relationship is delineated through a piecewise expression detailed in Table 1. In instances where the temperature falls beyond the range stipulated in the table, the corresponding minimum or maximum UTS value is employed for the analysis.

2.3. Implementation and verification of tritium migration model

2.3.1. Model theory of tritium migration in FeCrAl alloy

As shown in Fig. 1, the main processes of the tritium diffusion in UO_2–FeCrAl combination are explored:

① Calculate the tritium production rate in different reactor cores and the total amount of its diffusion into the gap (Q_s).

For a fuel with a linear power density of 220 W/cm, an estimated tritium generation rate is adopted from Katakura (2011) ($T\_product\_rate = 1.335\times10^9 \mathrm{atom/cm^3/s}$). According to Dollè, L. et al. (1980) recommendation, about 50% of the tritium produced is released into the gap. Therefore, the total amount of tritium diffused from the reactor core into the gap could be calculated by the following formula:

$$T_{total} = T\_product\_rate \times t \tag{11}$$

where t is the total calculation time with a value of 1.2×10^5 s based on work by Hu et al. (2015).

② Calculate the concentration of tritium diffused from the gap to the inner surface of the cladding. (Assuming that tritium is in local equilibrium at the gap and cladding interface).

$$J_{i\_c} = J_{g\_i} = aQ_{g\_c} = a\left(\frac{dT_g(t)}{dt} - Q_s + \lambda T_g(t)\right) \tag{12}$$

where Q_{g_c} is the absorption rate of tritium at the cladding inner surface and a is the ratio of the gap volume to the cladding inner surface area in m.

And the tritium flux from the cladding inner surface into the bulk is described by Fick's first law:

$$J_{i\_C} = -D_T\frac{\partial T_c(0,t)}{\partial x} \tag{13}$$

According to Sievert's law, the expression for the equilibrium tritium concentration on the cladding inner surface is:

$$T_c(0,t) = Sp^{\frac{1}{2}} = SaT_g^{\frac{1}{2}} \tag{14}$$

where S is the tritium solubility in the cladding material, p is the tritium partial pressure in the gap and a used to convert the tritium concentration in the gap to gas pressure is a coefficient derived from the ideal gas equation.

The first boundary condition (gap side) is concluded combing Eqs. (10)–(12) according to Hu et al. (2015), in the form of,

Table 1
UTS of FeCrAl cladding.

Temperature (K)	UTS (MPa)
294.738	569.475
551.495	543.205
644.048	527.023
829.85	288.826
1011.95	65.373

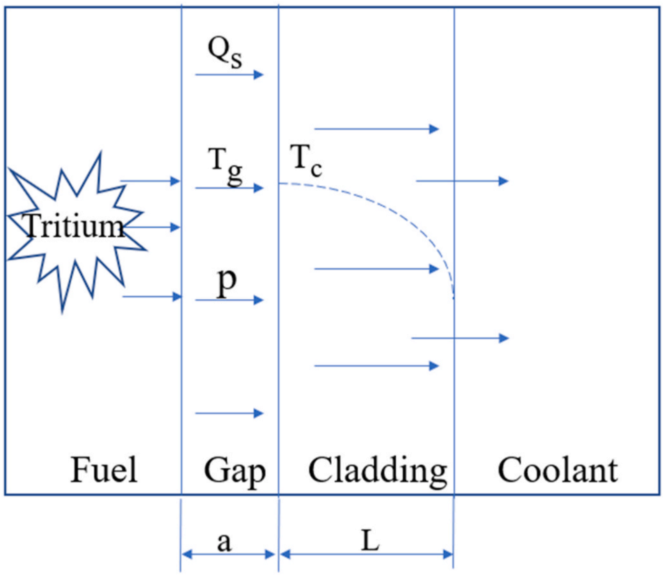

Fig. 1. The tritium migration process. Q_s is the hypothetical homogeneous tritium source released from fuel pellet. p is the tritium partial pressure in the gap. T_g is the tritium concentration in the gap. T_c is the tritium concentration in the cladding. a is the gap width. L is the cladding thickness.

$$2\alpha a T_c(0,t)\frac{\partial T_c(0,t)}{\partial t} - aQ_s + \alpha\lambda a(T_c(0,t))^2 = D_T\frac{\partial T_c(0,t)}{\partial x} \tag{15}$$

where $\alpha = \frac{1}{S^2 a^2}$.

The second boundary condition fixed zero concentration is set on the cladding outer surface (water side). And the cladding initial concentration is 0.

Based on Fick's law, the tritium diffusion in the cladding is calculated as:

$$\frac{\partial T_c(x,t)}{\partial t} = D_T\frac{\partial^2 T_c(x,t)}{\partial x^2} - \lambda T_c(x,t) \tag{16}$$

where λ is the decay constant of tritium and D_T is the tritium diffusion coefficient in FeCrAl alloy, which is mainly obtained by designing experiments to measure the hydrogen diffusion coefficient in FeCrAl alloy. Simultaneously, the hydrogen diffusion coefficient in FeCrAl alloy will be verified by first-principle calculation based on density functional theory. Subsequently, according to the principle that the diffusion coefficient of isotopic elements is inversely proportional to the square root of their mass, the tritium diffusion coefficient in FeCrAl alloy is computationally determined. The tritium diffusion coefficient in FeCrAl alloy is depicted in Fig. 2. It can be seen from the figure that the diffusion coefficient obtained by theoretical calculation is close to that measured by experiment. This alignment between theoretical calculation and experimental data demonstrated the robustness and accuracy of the employed calculation methodology

2.3.2. *Validation of tritium migration model in FeCrAl alloy*

Given that the fuel thermodynamic model has been verified in our previous work by Liu et al. (2016), the tritium migration model is mainly validated in this work.

Based on the above introduced model, the distribution of tritium concentration within the cladding is calculated by solving the tritium migration model through COMSOL. Subsequently, a comparative analysis is conducted involving the obtained results, reported data, and the modeling outcomes derived from BISON (Oneal,2021; Yin et al., 2022). The tritium concentration distribution within FeCrAl are forecasted in Figs. 3–5, evolved from a transient phase to a steady state. During the transient phase, accurate calculation of tritium concentration within the

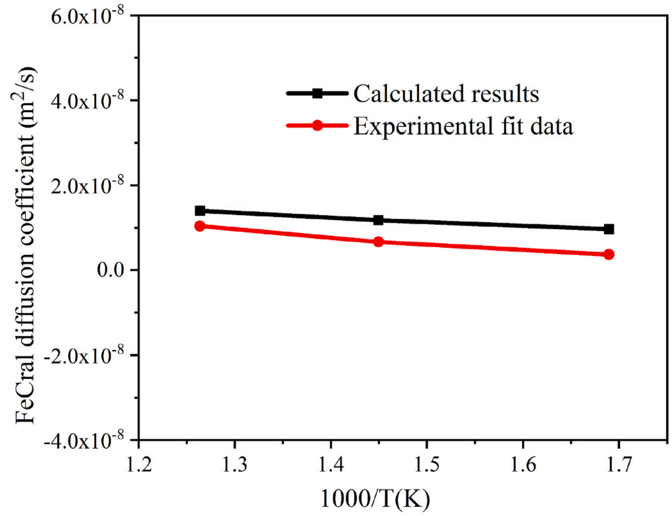

Fig. 2. Tritium diffusion coefficient in FeCrAl alloy.(Gao, 2022)

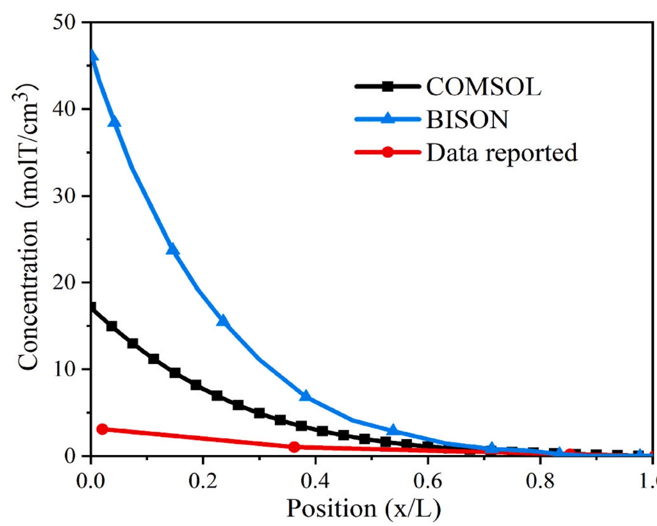

Fig. 3. Calculated results and reported experimental result of different models for the tritium distribution in FeCrAl cladding at 1000 s ($\times$ 10^{12} times) Oneal, 2021.

cladding is challenging, both BISON and COMSOL are exhibited discrepancies in comparison to the reported values. Nonetheless, in this transient phase, a higher degree of consistency and accuracy is demonstrated based on COMSOL in predicting tritium distribution trends within the cladding, as opposed to BISON. Besides, due to disparate diffusion coefficients employed in BISON and COMSOL respectively, the BISON and COMSOL results are shown differently. And because of the different diffusion coefficients being calculated and the chosen mesh settings, the COMSOL results are slightly larger in steady state. Overall, the accuracy of the tritium migration model developed based on COMSOL is validated from these results.

3. Results and discussions

In this section, the fuel performance of the UO_2 fuel pellet with three types of claddings including Zircaloy, Cr-coated Zircaloy and FeCrAl are compared separately under normal operating and accident conditions of PWR. Additionally, the tritium migration related claddings is discussed under fixed tritium concentration at the boundary.

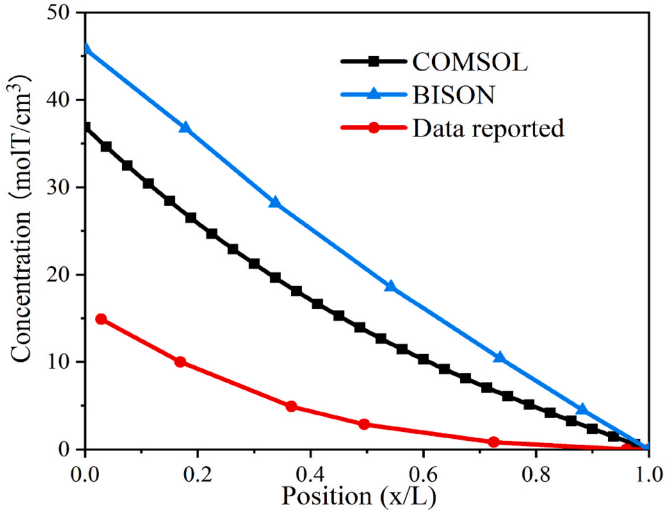

Fig. 4. Calculated results and reported experimental result of different models for the tritium distribution in FeCrAl cladding at 5000 s ($\times 10^{12}$ times) (Oneal, 2021).

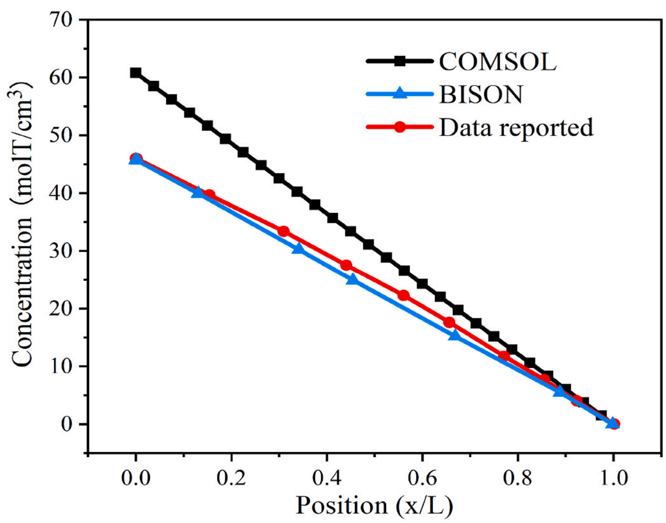

Fig. 5. Calculated results and reported experimental result of different models for the tritium distribution in FeCrAl cladding at 120000 s ($\times 10^{12}$ times) (Oneal, 2021).

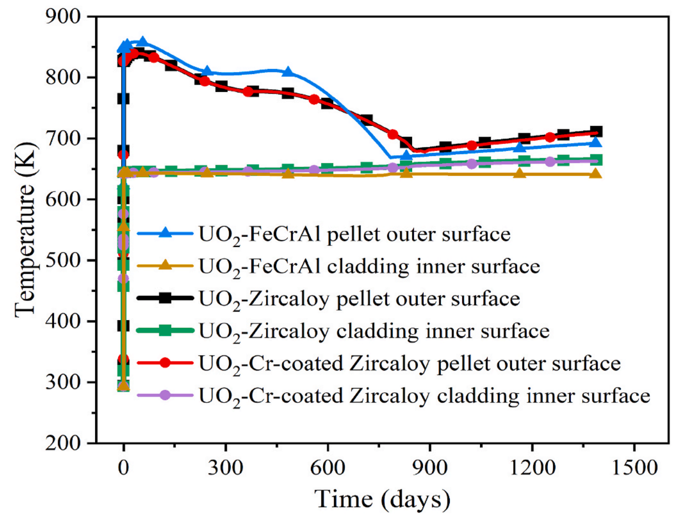

Fig. 6. The fuel pellet outer surface and cladding inner surface temperature evolution of different fuel and cladding combinations under normal operating condition.

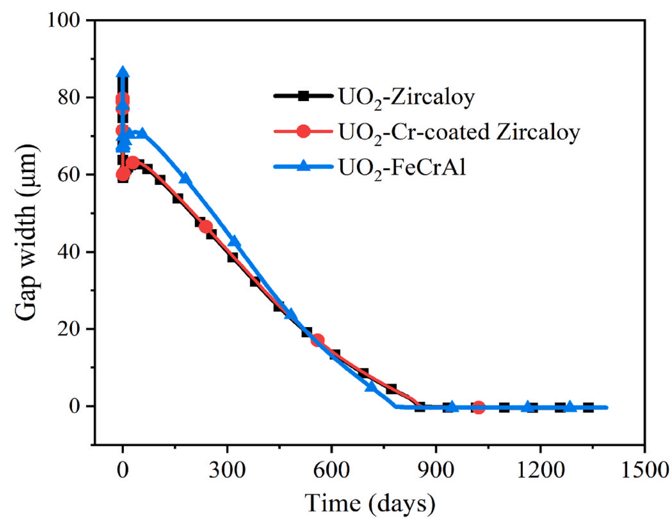

Fig. 7. Gap width evolution of different fuel and cladding combinations under normal operating condition.

3.1. Under normal operating condition

The temperature of fuel pellet outer surface and cladding inner surface for three types of fuel cladding combinations (UO_2-Zircaloy, UO_2–Cr-coated Zircaloy and UO_2–FeCrAl) under normal operating condition are depicted in Fig. 6. Notably, there is basically no difference in the temperature evolution between UO_2-Zircaloy and UO_2–Cr-coated Zircaloy due to the fact that the Cr-coated Zircaloy cladding is found to exhibit comparable thermo-mechanical performance to the un-coated Zircaloy cladding, and this is consistent with the findings of Wagih et al. (2018). At the beginning of the reaction, the fuel temperature is found to be rapidly increased due to the rapid power increase. Subsequently, the temperature-induced thermal expansion of the fuel causes the gap width to be rapidly reduced. The evolution of gap width between the fuel pellet and cladding under normal operating condition for three types of the fuel cladding combinations are illustrated in Fig. 7. During the period from 0 to 800 days, all three types of combinations are found to have a consistent decrease in outer surface temperature, which is ascribed to the continuous decrease of gap width. The decreased gap width causes the rise of gap thermal conductivity followed by the enhanced heat transfer ability of the fuel rod. And during this period, the fuel pellet outer surface temperature of the UO_2–FeCrAl is calculated to be higher than others before 700 days due to gap width influence. After 700 days, it turns to be lower due to gap width changes. This phenomenon occurs because a larger gap width results in a decline in the thermal conductivity of the gap, consequently leading to elevated temperature on the outer surface of the fuel pellet. From Figs. 6 and 7, the delayed changes of temperature and gap width are found because of the effects of gap fission gas. After about 800 days, the gap thermal conductivity drops, decreasing heat transfer and increasing the fuel pellet outer surface temperature. The related gap thermal conductivity parameters for the three types of fuel cladding combinations are detailed in Figs. 7–9. Furthermore, it's worth noting that in the UO_2–FeCrAl combination, both the outer surface temperature of the fuel pellet after gap closure and the cladding inner surface temperature are observed to remain lower compared to the other two combinations. This is attributed to the higher thermal conductivity of the FeCrAl cladding in contrast to the Zircaloy cladding, and the heat is easier to be conducted out of the cladding.

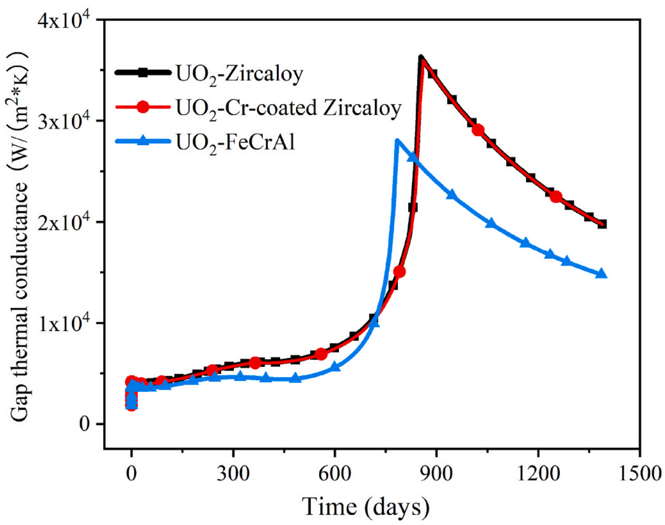

Fig. 8. Gap thermal conductance evolution of different fuel cladding combinations under normal operating condition.

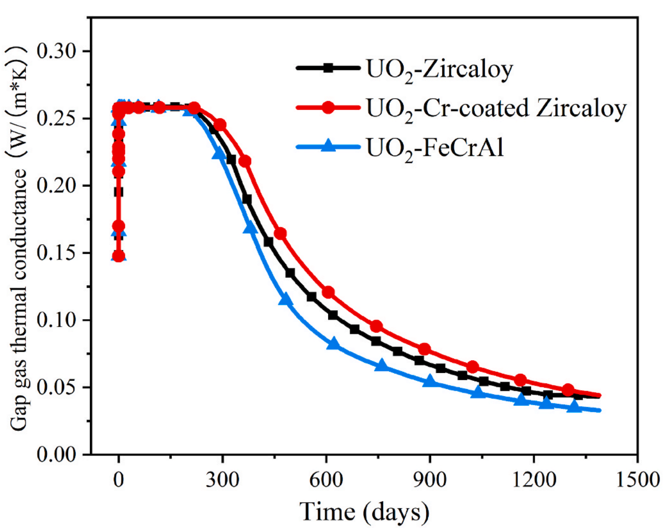

Fig. 9. Gap gas thermal conductance evolution of different fuel and cladding combinations under normal operating condition.

As depicted in Fig. 7, an increase in gap width is observed for a short period of time due to the shrinkage of the pellet size caused by fuel densification. Subsequently, due to concurrent factors of fission gas generation within the fuel and cladding creep-induced expansion, the gap width of different fuel cladding combinations under normal operating condition is found to decrease until complete closure is achieved. During the period from 0 to 600 days, the gap width of the UO_2–FeCrAl combination is observed to be significantly larger compared to the other two combinations, as a consequence, this leads to a lower gap thermal conductivity of the former compared to the latter. This thermal conductivity difference can be further demonstrated by examining the change of gap thermal conductivity of different fuel cladding combinations. As depicted in Fig. 8, between the interval approximately ranging from 700 to 800 days, the UO_2–FeCrAl combination exhibits heightened gap thermal conductance, which is due to its smaller gap width compared to the other two combinations. And we can see from Figs. 7 and 8, the delayed evolution time between the gap width and the gap thermal conductance can be observed due to the more predicted fission gas release. Furthermore, gap closure is found to occur first in the UO_2–FeCrAl combination in Fig. 7, which is the consequence of its greater thermal expansion coefficient. After the gap closure, the gap thermal conductivity is predominantly determined by the gas thermal conductivity, which experiences a sharp reduction as the gas thermal conductivity declines. Obviously, the gap thermal conductivity of UO_2–FeCrAl combination is found to peak first, and then the fuel pellet outer surface temperature is found to rise first at this time. The evolution of gas thermal conductivity for different fuel combinations is shown in Fig. 9, and a decreasing trend is observed for the gas thermal conductivity of the three types of fuel cladding combinations. Specifically, the gas thermal conductivity of UO_2–FeCrAl combination is calculated to be lower than that of UO_2-Zircaloy combination. So the gap thermal conductivity of different fuel combinations is found to decrease in Fig. 8 and the gap thermal conductivity of UO_2–FeCrAl combination is consistently lower than that of UO_2-Zircaloy combination after gap closure.

The evolutions of the fission gas release for the three types of fuel cladding combinations (UO_2-Zircaloy, UO_2–Cr-coated Zircaloy and UO_2–FeCrAl) under normal operating condition are compared in Fig. 10. The fission gas releases of all the combinations are found to be basically the same, where the tinny differences are mainly related to the fuel temperature. During the initial operating period, the UO_2–FeCrAl combination exhibits a greater fission gas release compared to the other two combinations. The mechanical contact between the pellet and the cladding will be aggravated because of the gaseous swelling caused by the fission gas accumulation. Consequently, this mechanical interaction leads to detrimental effects on the cladding, influencing the lifespan of the fuel rods. Simultaneously, heat transfer will be weakened due to the fission gas release, leading to an increase in the fuel temperature. The heightened fuel temperature, in turn, prompts a higher release of fission gases, causing the safety risks. However, it is noteworthy that as the operating time progresses, the UO_2–FeCrAl combination demonstrates a decreased fission gas release in comparison to the other two combinations. This reduction ultimately enhances the safety of reactor operations.

3.2. *Under LOCA condition*

The temperature evolution trends of fuel pellet outer surface and cladding inner surface for three types of fuel cladding combinations (UO_2-Zircaloy, UO_2– Cr-coated Zircaloy and UO_2–FeCrAl) under LOCA condition are illustrated in Fig. 11. From 0 s to 90 s, the temperature of different fuel cladding combinations is observed to increase gradually, followed by a downward trend after 90 s. This initial temperature rise is attributed to the rapid decrease in coolant surrounding the fuel at the

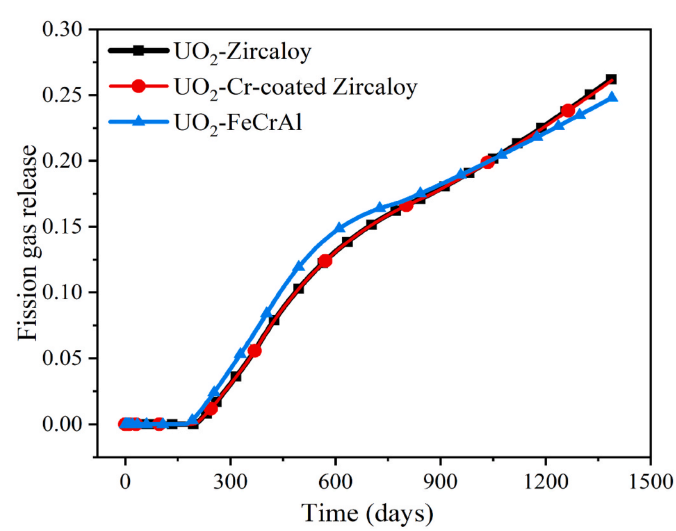

Fig. 10. Fission gas release evolution of different fuel and cladding combinations under normal operating condition.

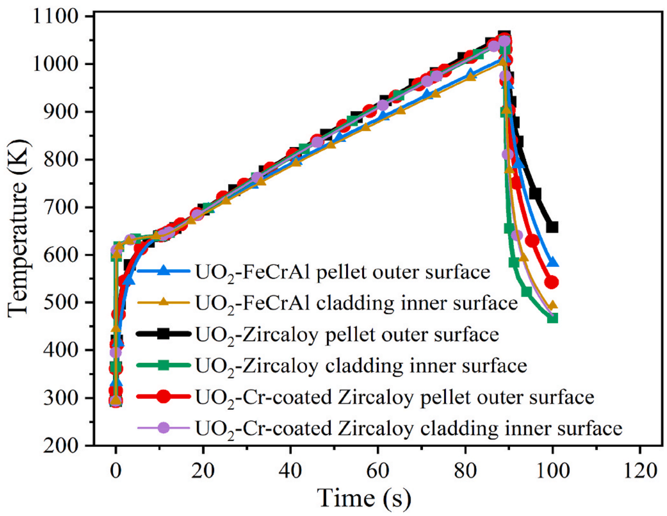

Fig. 11. The fuel pellet outer surface and cladding inner surface temperature evolution of different fuel and cladding combinations under LOCA condition.

onset of LOCA, leading to an interruption in the normal heat transfer process. Consequently, the fuel is in an adiabatic heating state, resulting the temperature of different fuel combinations increased obviously. However, at about 90 s, a notable shift occurs as the coolant within the sub-channels gets replenished, facilitating the resumption of proper heat transfer from the fuel to the surroundings. Consequently, the fuel temperatures begin to rapidly decrease. It is worth noting that throughout the LOCA event, the UO_2–FeCrAl combination consistently maintains lower temperatures in comparison to the other two combinations. variance can be attributed to the superior thermal conductivity of FeCrAl cladding relative to Zircaloy cladding. Higher thermal conductivity implies more effective heat transfer, leading to reduced temperatures. Additionally, the lower thermal diffusivity of FeCrAl is also one of the reasons for the lower temperature of the UO_2–FeCrAl combination. Thermal diffusivity is inversely influenced by density and heat capacity and directly influenced by thermal conductivity, the difference in density and heat capacity makes the thermal diffusivity lower. The thermal diffusivity for Zircaloy and FeCrAl claddings are compared in Fig. 12. The thermal diffusivity characterizes the material temperature increase rate. Higher thermal diffusivity denotes greater uniformity in internal temperature and a quicker material response to temperature

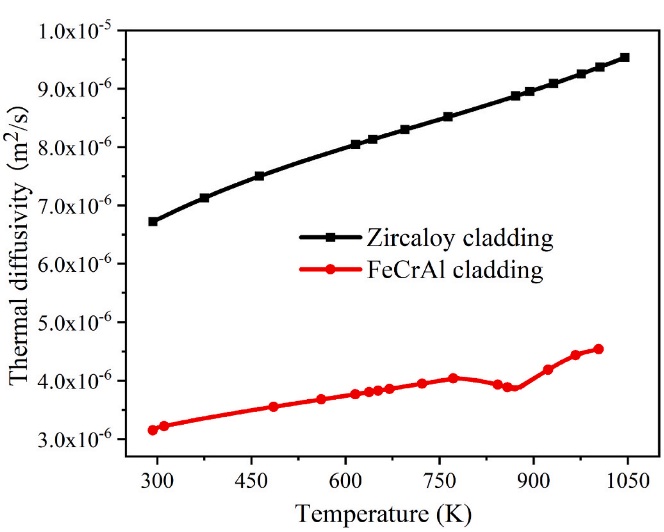

Fig. 12. Thermal diffusivity of Zircaloy and FeCrAl claddings under LOCA condition.

changes under transient conditions. As depicted in Fig. 12, the thermal diffusivity of the Zircaloy cladding is calculated to be higher than that of the FeCrAl cladding. Consequently, the UO_2-Zircaloy combination exhibits more pronounced temperature elevation compared to the UO_2–FeCrAl combination. Therefore, the superior temperature performance of the FeCrAl cladding over the Zircaloy cladding is derived under LOCA condition.

Table 2 presents the failure time for different fuel cladding combinations under LOCA condition. Two types of combinations (UO_2–Cr-coated Zircaloy and UO_2–FeCrAl) are found not to fail during the whole accident process, while the UO_2-Zircaloy combination is found to fail and to burst at 82 s. This is attributed to the higher Young's modulus and enhanced stiffness of the Cr-coated material compared to Zircaloy. The mechanical properties of entire cladding are improved following the application of a chromium (Cr) coating onto the Zircaloy cladding. Furthermore, the absence of failure in the case of FeCrAl cladding is attributed to the consistently lower hoop stress relative to the failure stress, as depicted in Fig. 13. As a result, the cladding failure time is delayed by applying the combinations of Cr-coated Zircaloy and FeCrAl cladding and the fuel safety is improved under LOCA condition. Importantly, the incorporation of Cr-coated Zircaloy cladding and FeCrAl cladding also yields enhanced fuel properties concerning oxidation and corrosion resistance when contrasted with uncoated Zircaloy cladding. This improvement contributes to the overall stability and safety of the fuel under LOCA condition.

3.3. Under RIA condition

The temperature evolution trends of fuel pellet outer surface and cladding inner surface for three types of fuel cladding combinations (UO_2-Zircaloy, UO_2–Cr-coated Zircaloy and UO_2–FeCrAl) under the RIA condition are compared in Fig. 14. Since the fuel pellet power changes abruptly under RIA condition, the temperature can be observed to increase suddenly and steeply at 0.06 s from the figure, and it returns to be normal at 0.15 s. Obviously, the thermodynamic behavior of UO_2-Zircaloy and UO_2–Cr-coated Zircaloy are found to be basically the same. Meanwhile, the fuel pellet outer surface temperature of UO_2–FeCrAl combination can be seen to be higher than that of the other two combinations from the figure. This discrepancy can be attributed to the substantial and rapid heat release from the fuel pellet under RIA accident condition. Additionally, the gap thermal conductivity of UO_2–FeCrAl combination is comparatively lower, impeding efficient heat transfer from the gap in a short period of time. Furthermore, the inner surface temperature of the cladding within the UO_2–FeCrAl combination remains lower than that of the other two combinations. This phenomenon arises due to the higher thermal conductivity of the FeCrAl alloy compared to the counterparts. Consequently, the cladding effectively exports heat, contributing to the temperature moderation observed in this scenario.

The cladding outer surface total strain evolution of the UO_2-Zircaloy, UO_2–Cr-coated Zircaloy and UO_2–FeCrAl fuel combinations under the RIA condition are illustrated in Fig. 15. It can be observed from the figure that UO_2–FeCrAl combination exhibits a larger total strain compared to the other two combinations. The cladding's total strain is considered as the sum of thermal strain and cladding creep in CAMPUS settings. And the cladding creep is found to have almost no change in a very short period of time. However, it is evident that the thermal strain of the FeCrAl cladding is greater than that of the other two claddings, resulting in the overall greater total strain for the UO_2–FeCrAl combination. The cladding outer surface thermal strain evolution is depicted

Table 2
Calculated value of fuel failure time.

Fuel combination	UO_2-Zircaloy	UO_2-Zircaloy with Cr coating	UO_2–FeCrAl
failure time (s)	82	No failure	No failure

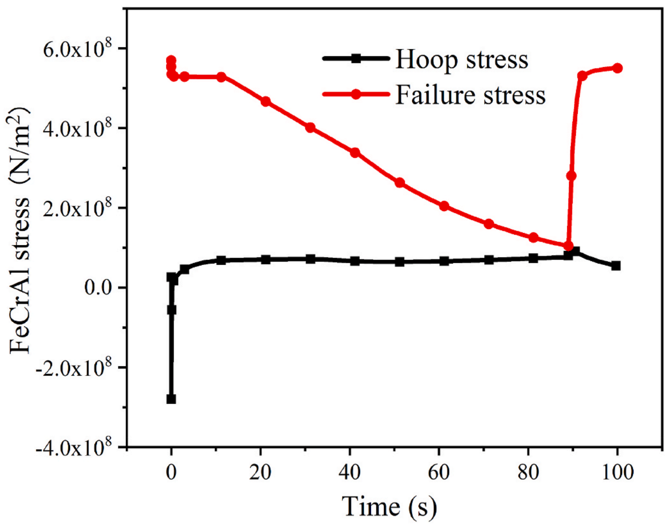

Fig. 13. Stress evolution of FeCrAl cladding under LOCA condition.

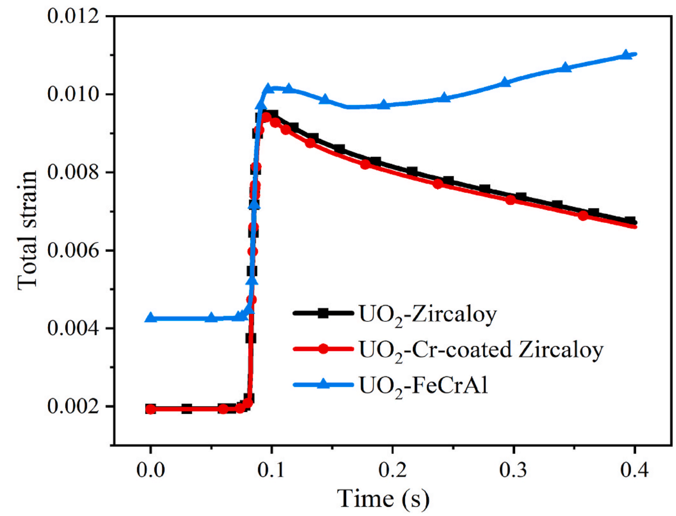

Fig. 15. Cladding outer surface total strain evolution of different fuel and cladding combinations under RIA condition.

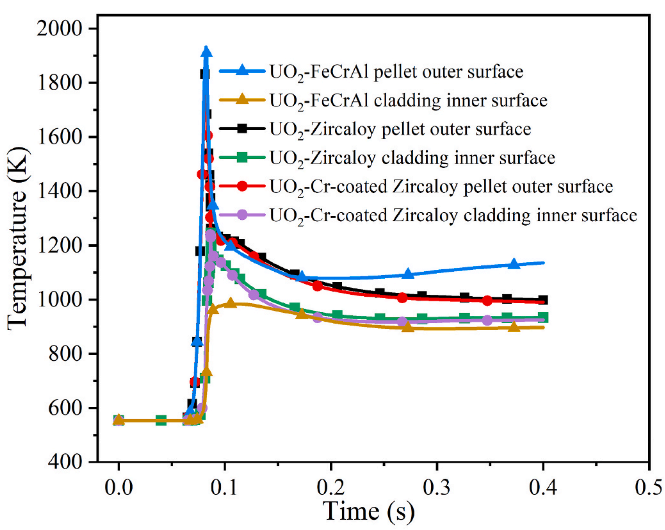

Fig. 14. The fuel pellet outer surface and cladding inner surface temperature evolution of different fuel and cladding combinations under RIA condition.

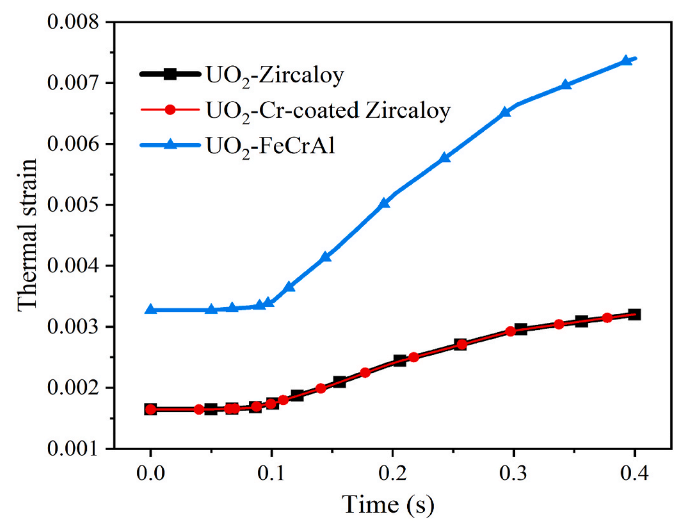

Fig. 16. Cladding outer surface thermal strain evolution of different fuel and cladding combinations under RIA condition.

in Fig. 16. As we know, the large strain is found to lead to the cladding deformation easily, which is not conducive to the normal operating condition. Therefore, the mechanical properties of FeCrAl cladding appear to be comparatively less favorable than those of the other two cladding materials under RIA condition, as indicated through the aforementioned discussions.

Based on the comprehensive analysis conducted above, a clear improvement in the mechanical performance of Zircaloy cladding is evident when it is coated with chromium (Cr) within a PWR environment. This improvement extends to the cladding's oxidation resistance and corrosion resistance, enhancing its resilience both normal operating and accident conditions. That is to say, the adoption of Cr-coated Zircaloy cladding demonstrates a promising potential as an ATF cladding solution. Conversely, FeCrAl cladding is reported to have advantages in terms of thermal hydraulic performance under normal and LOCA conditions. However, it is important to note a potential drawback associated with FeCrAl cladding, specifically, an accelerated propensity for tritium penetration, which needs to be further analyzed and evaluated. Subsequently, an examination of tritium penetration across different cladding materials is presented below.

3.4. *Tritium migration*

In order to facilitate a comparative evaluation of tritium penetration among different cladding materials, a fixed concentration boundary condition of 5×10^{-5} mol/m^2 is established on the left side of cladding (the gap side), as recommended by Hu et al. (2015). Tritium flux, coupled with temperature data under normal operating condition, is calculated based on the built-in module of COMSOL. Fig. 17 depicts the tritium flux permeating from different cladding into coolant. A significant observation is that the tritium diffusion rate within Zircaloy cladding experiences acceleration around 900 days, in contrast to the constant diffusion rate maintained by FeCrAl cladding. Because the temperature of the FeCrAl cladding in Fig. 6 does not fluctuate much during this period, the tritium diffusion rate in the cladding remains almost constant, so the tritium flux in FeCrAl cladding shown in Fig. 17 remains stable. Conversely, the temperature of the Zircaloy cladding tends to increase at this point, augmenting the cladding's tritium diffusion capacity and correspondingly elevating tritium flux. Furthermore, the output tritium flux of FeCrAl cladding is observed to stabilize quickly within a short time, preceding the stabilization of output tritium flux from Zircaloy cladding and Cr-coated Zircaloy cladding. This

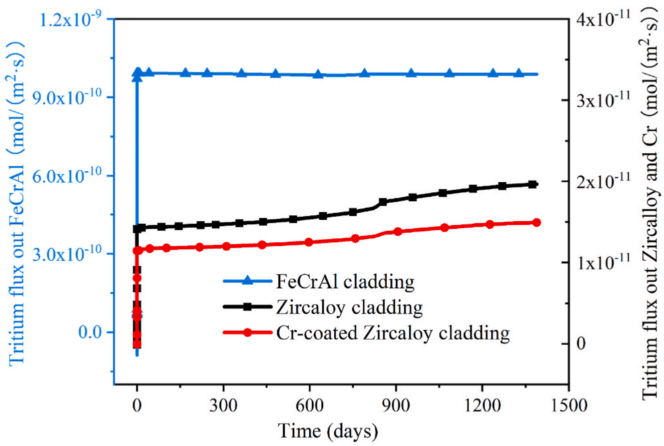

Fig. 17. Output tritium flux on the right side of different claddings (The left side of the ordinate axis represents the tritium flux of FeCrAl cladding. The right side of the ordinate axis represents the tritium flux of Zircaloy cladding and Cr-coated Zircaloy cladding).

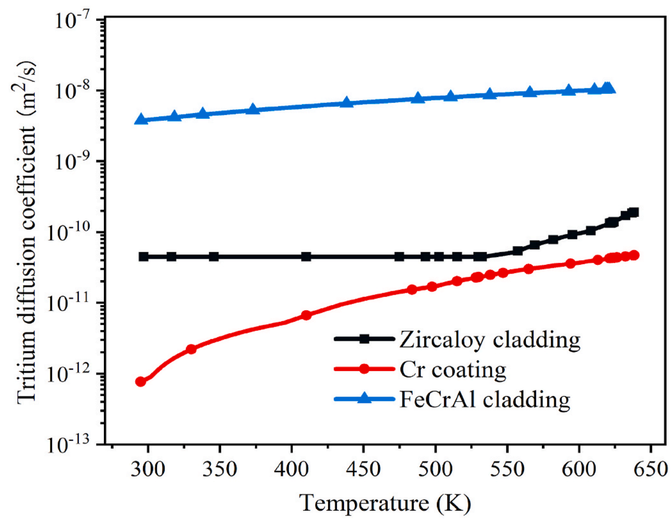

Fig. 18. The tritium diffusion coefficient of different materials.

behavior arises because the tritium diffusion coefficient of FeCrAl cladding significantly surpasses that of Zircaloy cladding by approximately two orders of magnitude. At the same time, it is evident that the output tritium flux on the right side of the FeCrAl cladding, Zircaloy and Cr-coated Zircaloy is sequentially reduced. The cumulative amount of tritium released from the different claddings to the coolant at 1388 days is summarized in Table 3. Notably, the total tritium release from Zircaloy cladding into the coolant during this interval is considerably lower than that from FeCrAl cladding, a contrast that becomes even more pronounced after the Zircaloy cladding is further coated with Cr coating. This discrepancy can be primarily attributed to the pivotal role of the tritium diffusion coefficient in influencing the flux in and out of the cladding. Greater cladding diffusion coefficients correspond to heightened tritium influx and efflux rates. The tritium diffusion coefficient of different claddings is depicted in Fig. 18. The diffusion coefficient of Zircaloy cladding is found to be smaller than that of FeCrAl cladding, and correspondingly, the diffusion coefficient of Cr coating is lesser than that of Zircaloy cladding. Consequently, Zircaloy cladding exhibits a reduction in output tritium flux compared to FeCrAl cladding. Notably, the introduction of a chromium layer onto the outer surface of Zircaloy cladding proves effective in further diminishing the total tritium release from the cladding into the coolant, thereby enhancing its tritium resistance characteristics.

4. Conclusions

As discussed above, the fuel performance of UO_2 fuel pellet combined with Zircaloy cladding, Cr-coated Zircaloy cladding, and FeCrAl cladding have been investigated under normal operating and accident conditions respectively. Additionally, further calculation of tritium migration is conducted for different claddings under normal operating condition. The conclusions are summarized below based on the investigation of the fuel performance and tritium resistance performance of the studied cases:

Table 3
Cumulative amount of tritium released from different claddings to the coolant at 1388 days.

fuel combination	UO_2-Zircaloy	UO_2-Zircaloy with Cr coating	UO_2–FeCrAl
Tritium accumulation (mol/m^2)	2.4×10^{-3}	1.8×10^{-3}	11.9×10^{-2}

1. Under normal operating condition, the thermodynamic performance of the fuel pellet equipped with the Cr-coated Zircaloy cladding and Zircaloy cladding is found to be basically the same. Notably, the primary objective of applying a Cr coating lies in preventing direct contact between Zircaloy and the coolant, thereby mitigating cladding corrosion and oxidation. Furthermore, the fuel pellet outer surface and cladding inner surface temperature of UO_2–FeCrAl combination are found to be lower than temperature of UO_2-Zircaloy combination. This distinction signifies the enhanced thermal performance of the UO_2–FeCrAl combination, fostering an improved reactor safety margin.
2. Under LOCA condition, the UO_2–FeCrAl combination is found to exhibit better temperature characteristics than UO_2-Zircaloy combination. Both the UO_2–Cr-coated Zircaloy combination and the UO_2–FeCrAl combination exhibit delayed cladding failure under LOCA conditions, thereby contributing to enhanced reactor safety.
3. Under RIA condition, the temperature performance of UO_2–FeCrAl combination is comparatively inferior to that of the UO_2–Cr-coated Zircaloy combination. Moreover, the UO_2–FeCrAl combination displays larger strain values in comparison to the UO_2–Cr-coated Zircaloy combination under RIA condition, which could potentially pose risks to reactor operation.
4. In terms of the tritium migration, the non-oxidized FeCrAl cladding is found to have the risk of accelerated the tritium migration rates, while the tritium flux and the amount of tritium diffused into the coolant can be decreased by using the UO_2-Zircaloy combination. And the tritium resistance performance is further improved when Zircaloy cladding is coated with Cr.

In summary, the UO_2–Cr-coated Zircaloy combination is found to be with great potential of accident tolerance. It is found to not only exhibit good thermodynamic performance, but also show substantial influence on tritium resistance. Importantly, the crack of Cr coating is not considered in this work, and the coupled simulation of real-time tritium generation within the fuel pellet and tritium diffusion within the cladding needs to be further considered, which will be the goal of our future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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